

Coloring, Covering and Partitioning

Proper Coloring:

Painting all the vertices of a graph with colors such that no adjacent vertices have same color, is called *proper coloring*.

A graph in which every vertex has been assigned a color according to proper coloring is called *properly colored graph*.

A graph can be properly colored in many different ways.

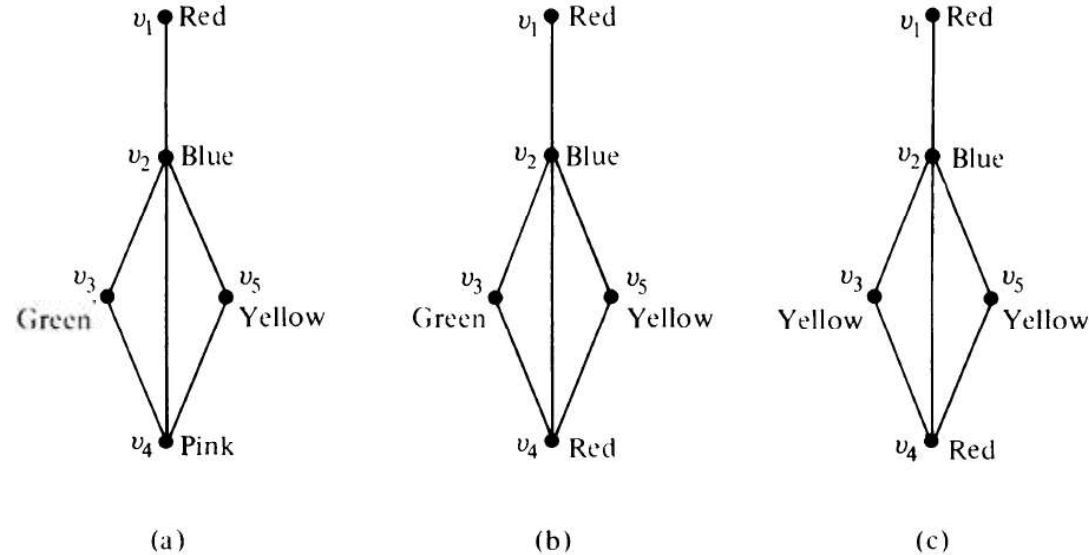


Fig. Proper colorings of a graph.

In Fig. (a), (b), (c) number of colors that has been used for proper coloring are 5,4,3 respectively. Hence the proper coloring is one that requires the minimum number of colors.

Chromatic Number:

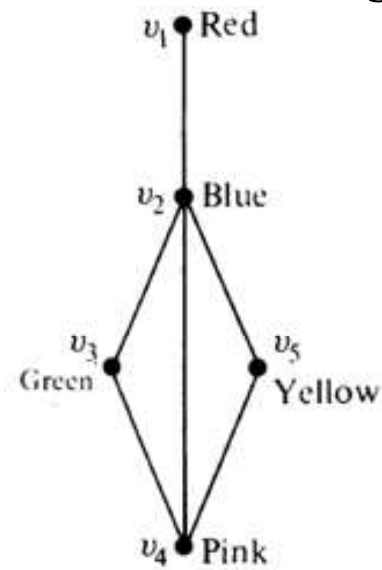
The least number of different colors required for proper coloring of the graph is called Chromatic number of the graph. Chromatic number of a graph G is denoted by $\chi(G)$.

If the graph requires κ *different colors* for its proper coloring and no less, then κ is called chromatic number. i.e. $\chi(G) = \kappa$.

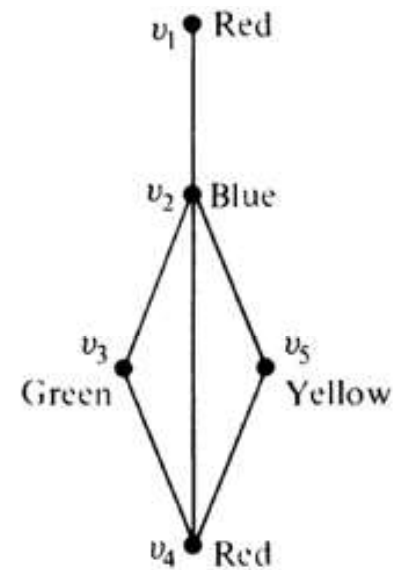
Chromatic graph:

The graph having chromatic number κ is called κ -Chromatic graph.

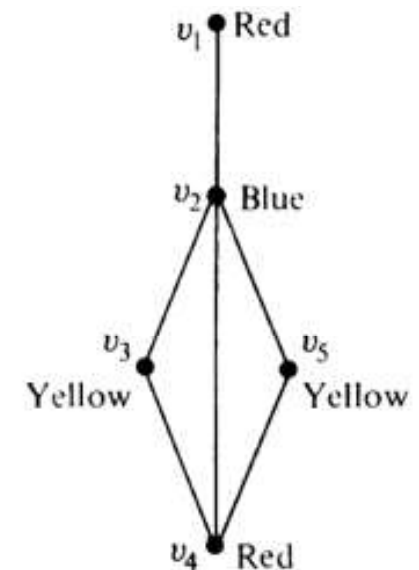
Ex: Given graph is 3-Chromatic graph.



(a)

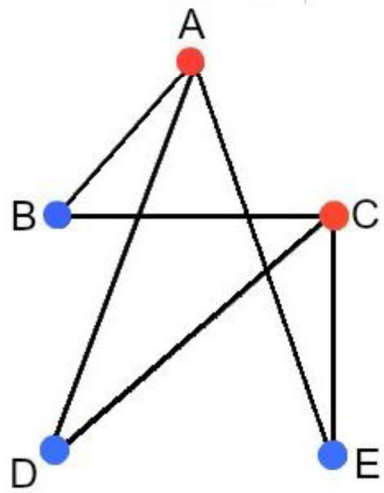


(b)

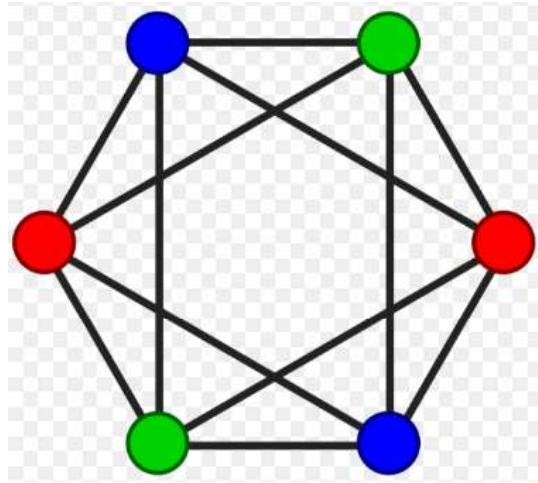


(c)

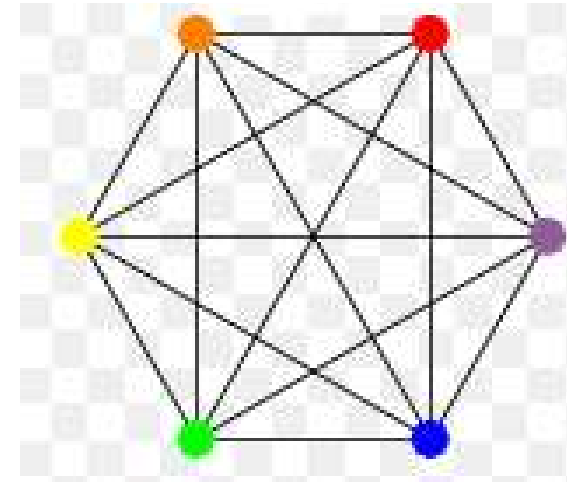
Fig. Proper colorings of a graph.



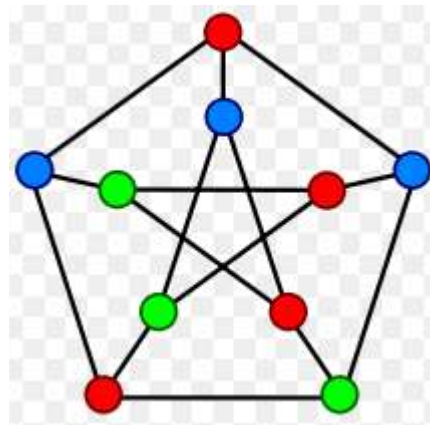
$\chi(G) = 2$
2-chromatic graph



$\chi(G) = 3$
3-chromatic graph



$\chi(G) = 6$
6-chromatic graph



$\chi(G) = 4$
4-chromatic graph

- Some observation from Proper Coloring of graphs:

1. A graph consist of only one vertex is 1-Chromatic.
2. A graph consist of one or more edges(not self loop) is at least 2-Chromatic.
3. A complete graph with n vertices is n -Chromatic as all its vertices are adjacent.
4. A graph containing a complete graph of r vertices is at least r -Chromatic. For ex: every graph containing a triangle is at least 3-Chromatic.
5. A graph consisting of simply one circuit with $n \geq 3$ vertices is **2-Chromatic, if n is even** and **3-Chromatic, if n is odd.**

Theorem: Every tree with two or more vertices is 2-chromatic. Is converse true.

Proof: Let T be a tree with root node v . Paint v with color 1. Paint all the vertices adjacent to v with color 2.

Next, paint the vertices adjacent to these (those that just have been colored with 2) using color 1. Continue this process till every vertex in T has been painted. Now in T we find that all vertices at odd distances from v have color 2, while v and vertices at even distances from v have color 1.

Now along any path in T the vertices are of alternating colors. Since there is one and only path between every pair of vertices in a tree, no two adjacent vertices have same color.

Thus T has been properly colored with two colors.

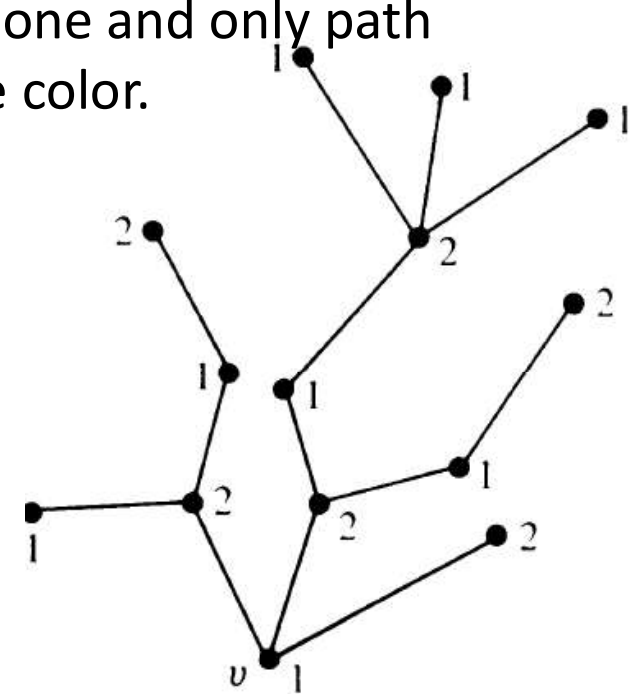
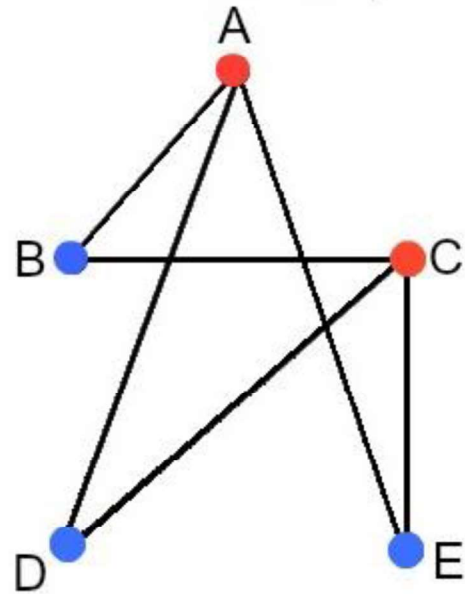


Fig. Proper coloring of a tree.

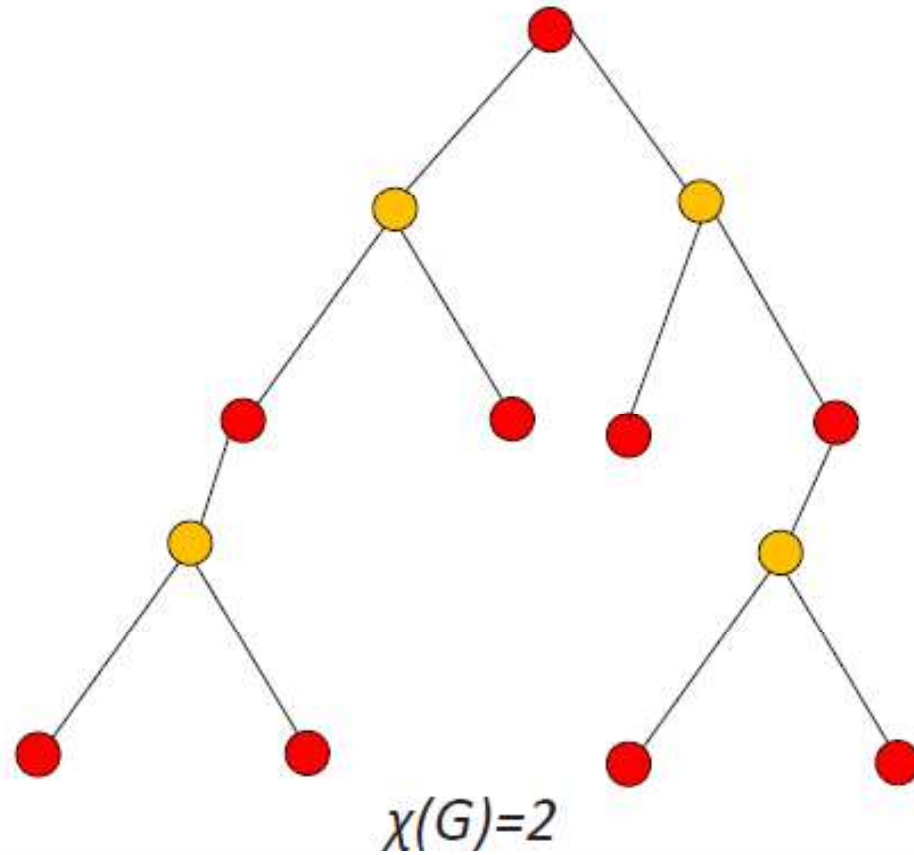
Every tree is 2-chromatic but not every 2-chromatic graph is tree.

Given graph is 2-chromatic but its not a tree.



$\chi(G) = 2$
2-chromatic graph

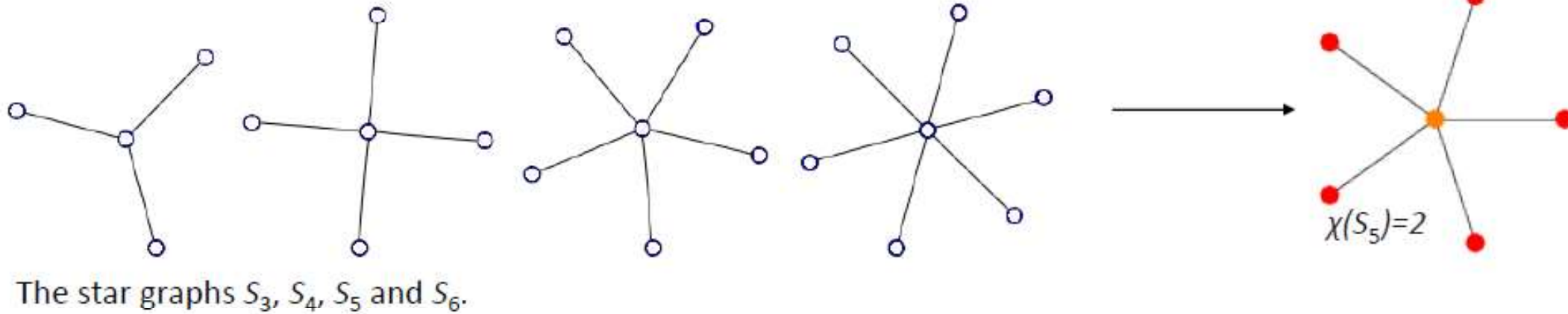
Chromatic Number of tree



So Chromatic Number of tree is 2

Chromatic number of Star Graph

- A star S_k is the complete bipartite graph $K_{1,k}$: a tree with one internal node and k leaves.



For all star graph chromatic number is: $\chi(G)=2$

THEOREM

A graph with at least one edge is 2-chromatic if and only if it has no circuits of odd length.

THEOREM

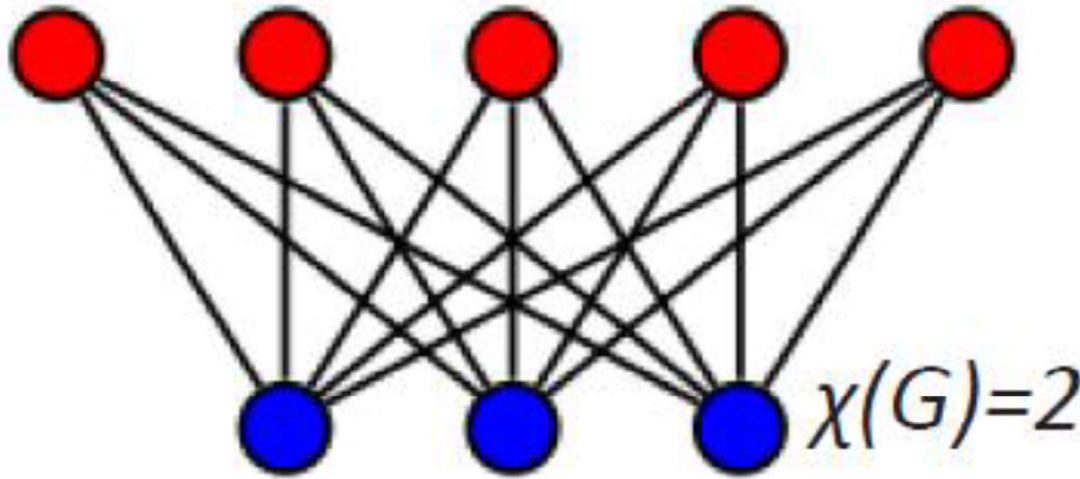
If $d_{\max}$ is the maximum degree of the vertices in a graph G ,

$$\text{chromatic number of } G \leq 1 + d_{\max}.$$

Chromatic Number of Bipartite Graph

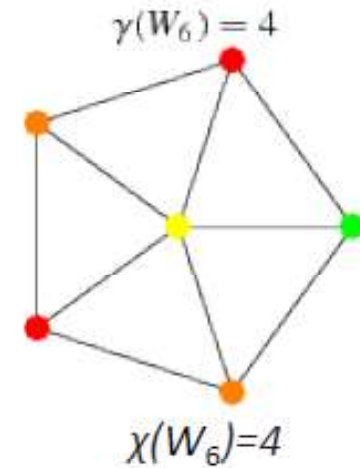
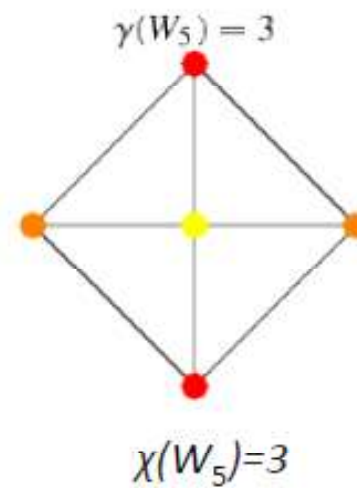
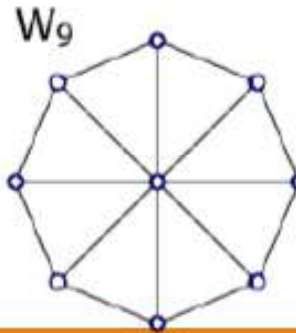
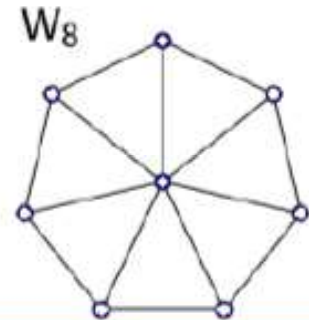
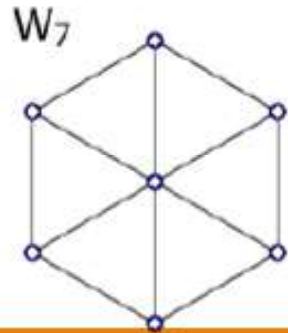
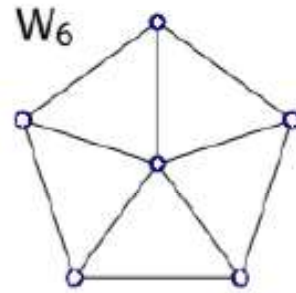
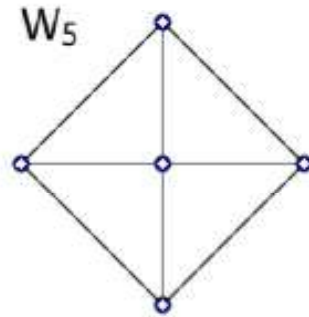
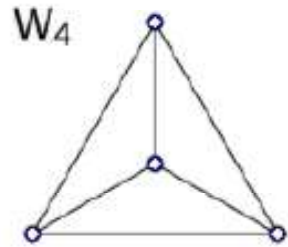
- A bipartite graph, also called a bigraph, is a set of graph vertices decomposed into two disjoint sets such that no two graph vertices within the same set are adjacent.

Clearly, every 2-chromatic graph is bipartite because the coloring partitions the vertex set into two subsets V_1 and V_2 such that no two vertices in V_1 (or V_2) are adjacent. Similarly, every bipartite graph is 2-chromatic.



Chromatic Number of Wheel Graph

- A wheel graph is a graph formed by connecting a single vertex to all vertices of a cycle.



A proper coloring of a graph naturally induces a partitioning of the vertices into different subsets. For example, the coloring in Fig. 1 produces the partitioning $\{v_1, v_4\}$, $\{v_2\}$, and $\{v_3, v_5\}$.

No two vertices in any of these three subsets are adjacent. Such a subset of vertices is called an independent set.

Independent set:

A set of vertices in a graph is said to be an *independent set* of vertices or simply an *independent set* if no two vertices in the set are adjacent.

$\{a, e\}$, $\{a, c, d\}$, $\{a, c, g\}$, $\{a, d, f\}$, $\{b, f\}$, $\{b, g\}$ etc are independent sets.

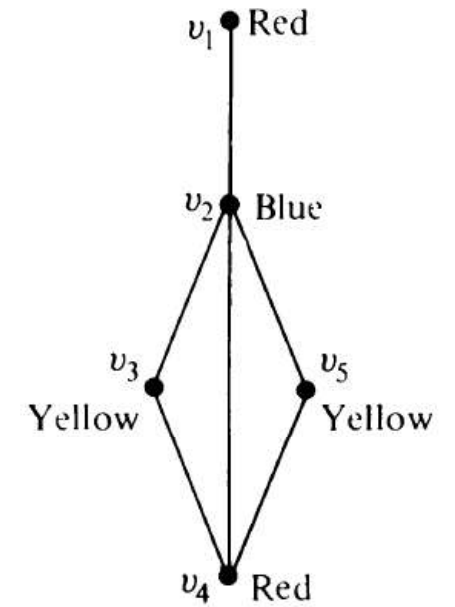


Fig.1

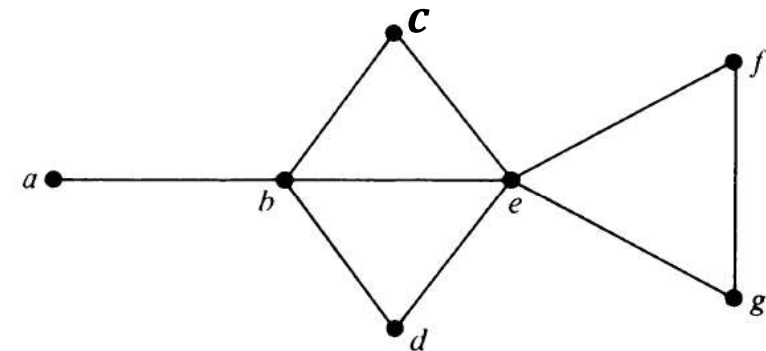


Fig. 2

Maximal Independent Set:

A *maximal independent set* (or *maximal internally stable set*) is an independent set to which no other vertex can be added without destroying its independence property. The set $\{a, c, d, f\}$ in Fig. is a maximal independent set. The set $\{b, f\}$ is another maximal independent set. The set $\{b, g\}$ is a third one.

From the example, it is clear that a graph, in general, has many maximal independent sets; and they may be of different sizes.

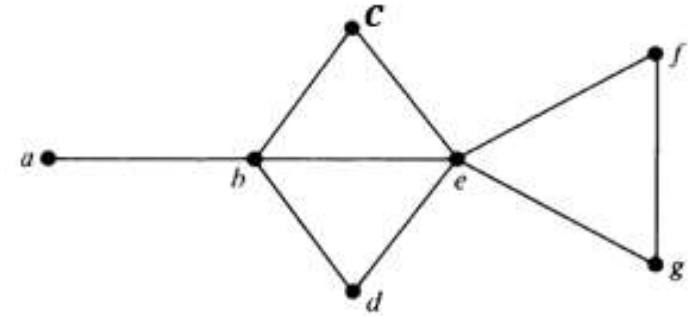


Fig. 2

Largest Independent Set:

A maximal independent set with largest number of vertices is called largest independent set.

In the given graph, the largest independent sets are $\{a, c, d, f\}$, $\{a, c, d, g\}$.

Independence Number:

The number of vertices in the largest independent set of a graph is called the independence number (or coefficient of internal stability), denoted by $\beta(G)$.

The largest independent sets are $\{a, c, d, f\}, \{a, c, d, g\}$.

Hence $\beta(G)=4$.

The largest independent sets are $\{a, c, d, f\}, \{a, c, d, g\}$.

Hence $\beta(G)=4$.

Note:

Consider a κ -chromatic graph G of n vertices properly colored with κ different colors. Since the largest number of vertices in G with the same color cannot exceed the independence number $\beta(G)$, we have the inequality

$$\beta(G) \geq \frac{n}{\kappa}.$$

Note: 1. While using proper coloring of a graph the vertex set with same color of vertices is also an independent set.

2. To find chromatic number of graph G , we must find the minimum number of maximal independent sets, which collectively include all the vertices of G .

Ex: In the given graph $\{a, c, d, f\}$, $\{b, g\}$, $\{a, e\}$ are containing all the vertices of G . Thus the graph is three chromatic.

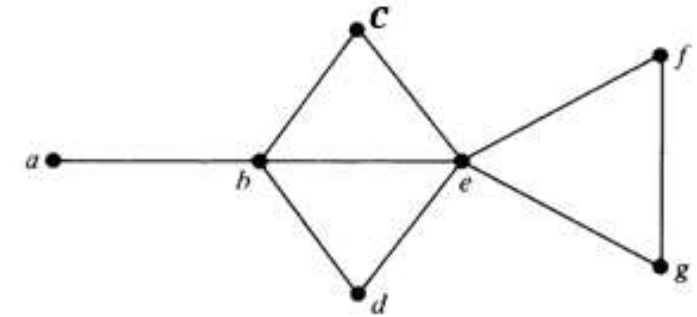


Fig. 2

Chromatic Partitioning:

Given a simple, connected graph G , partition all vertices of G into the smallest possible number of disjoint, independent sets. This problem, known as the *chromatic partitioning* of graphs, is perhaps the most important problem in partitioning of graphs.

The following are some chromatic partitioning of graph G

$\{(a, c, d, f), (b, g), (e)\},$

$\{(a, c, d, g), (b, f), (e)\},$

$\{(c, d, f), (b, g), (a, e)\},$

$\{(c, d, g), (b, f), (a, e)\}.$

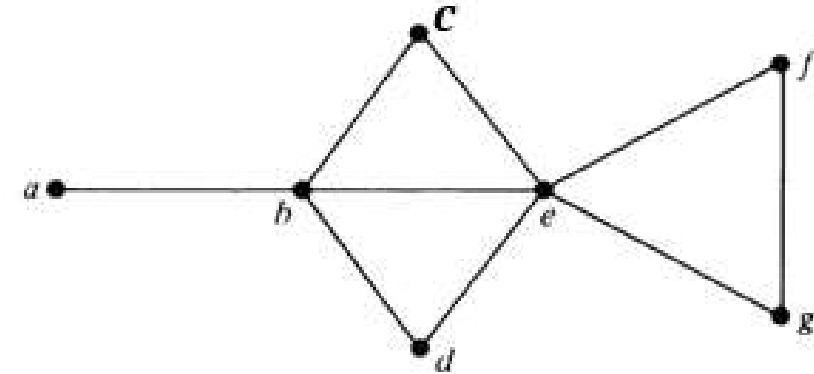
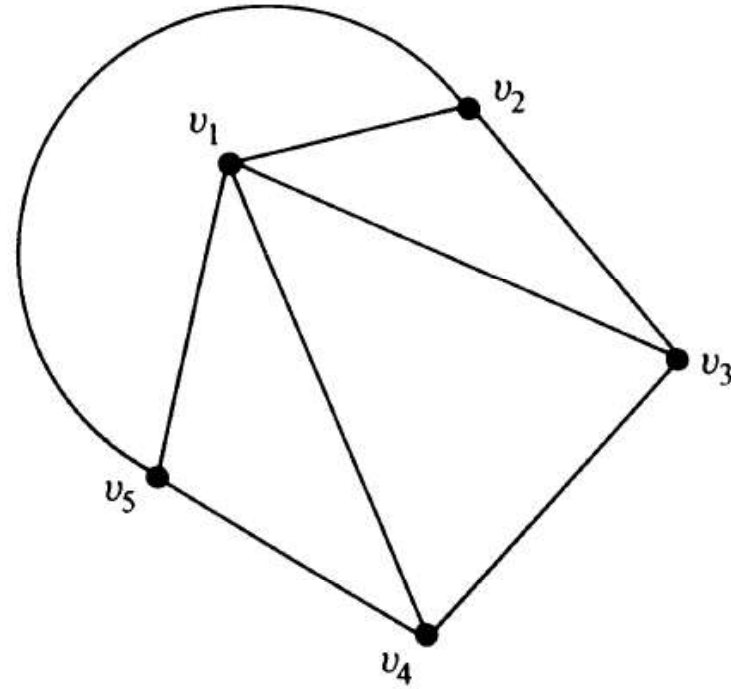


Fig. 2

The above graph doesn't have unique chromatic partitioning.

Uniquely Colorable Graphs: A graph that has only one chromatic partition is called a *uniquely colorable* graph.



A 3-chromatic graph.

The given graph has only one (unique) chromatic partitioning
 $\{v_1\}, \{v_2, v_4\}, \{v_3, v_5\}$

Fusion: A pair of vertices a, b in a graph are said to be *fused* (*merged* or *identified*) if the two vertices are replaced by a single new vertex such that every edge that was incident on either a or b or on both is incident on the new vertex. Thus fusion of two vertices does not alter the number of edges, but it reduces the number of vertices by one. See Fig. 2-16 for an example.

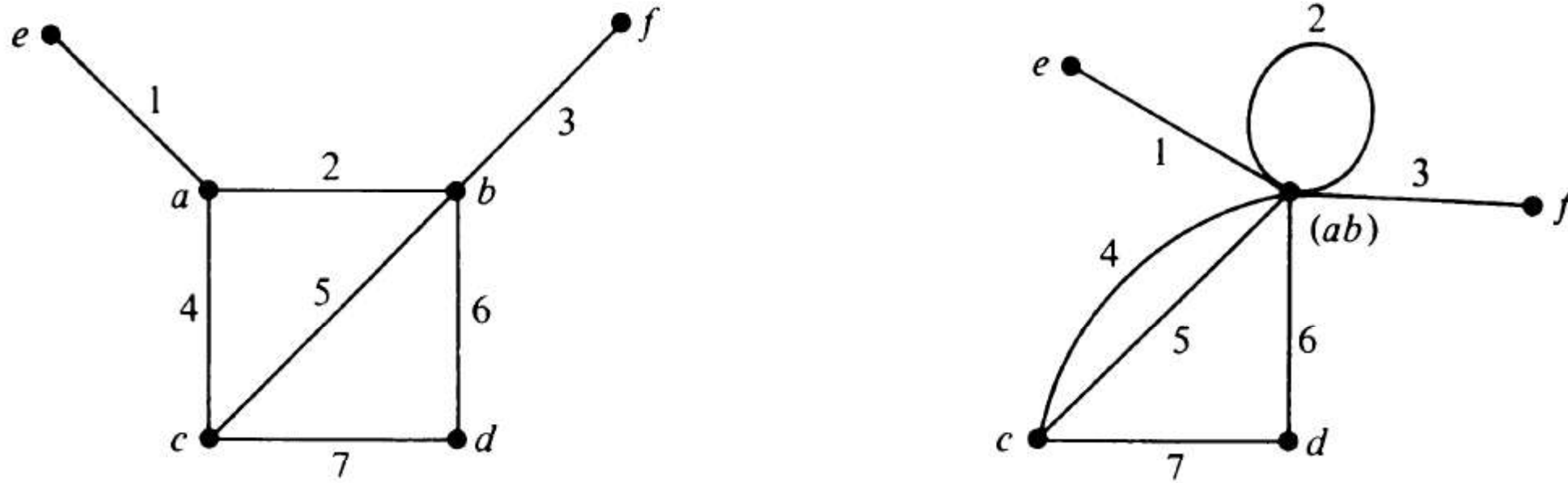


Fig. 2-16 Fusion of vertices a and b .

Chromatic polynomial

A given graph G of n vertices can be properly colored in many different ways using a sufficiently large number of colors. This property of graph can be expressed by means of a polynomial. This polynomial is called as the chromatic polynomial of G and is defined as follows:

The value of the chromatic polynomial $P_n(\lambda)$ of a graph with n vertices gives the number of ways of properly coloring G using λ or fewer colors.

$P_n(\lambda) =$ *The number of ways of proper coloring of a graph with n vertices, using λ colors*

Let c_i be the different ways of properly coloring graph G using exactly i colors.

Since i colors can be chosen out of λ colors in $\binom{\lambda}{i}$ different ways

No. of ways of proper coloring of G exactly i colors out of $\lambda = c_i \binom{\lambda}{i}$

Since i can be any positive integer from 1 to n (it is not possible to use more than n colors on n vertices), the chromatic number is the sum of these terms, that is

$$\begin{aligned} P_n(\lambda) &= \sum_{i=1}^n c_i \binom{\lambda}{i} \\ &= c_1 \frac{\lambda}{1!} + c_2 \frac{\lambda(\lambda-1)}{2!} + c_2 \frac{\lambda(\lambda-1)(\lambda-2)}{3!} + \dots \\ &\quad + c_n \frac{\lambda(\lambda-1)(\lambda-2) \dots (\lambda-n+1)}{n!} \end{aligned}$$

Each c_i has to be evaluated individually for the given graph.

For example, any graph with one edge requires atleast two colors for proper coloring and therefore

$$c_1 = 0.$$

A graph with n vertices and using n different colors can be properly colored in $n!$ ways, i.e.,

$$c_n = n!$$

Chromatic Polynomial of a complete graph:

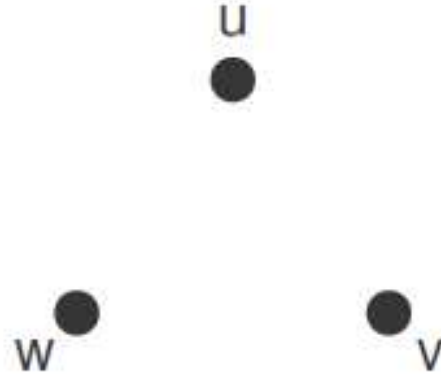
Number of ways of colorings properly of a complete graph with n vertices and λ colors is called a chromatic polynomial and it is denoted by

$$P_n(\lambda) = \lambda(\lambda - 1)(\lambda - 2) \dots (\lambda - (n - 1))$$

Properties of chromatic polynomial:

1. Degree of $P_n(\lambda)$ is n i.e. equal to the number of vertices.
2. Coefficient of λ^n is one i.e. chromatic polynomial is monic.
3. The constant term of $P_n(\lambda)$ is zero.
4. The coefficient of $|\lambda^{n-1}|$ is equal to the number of edges in G .
5. The coefficient of λ in disconnected graph is always zero.
5. If the graph is connected the coefficient of λ is nonzero.

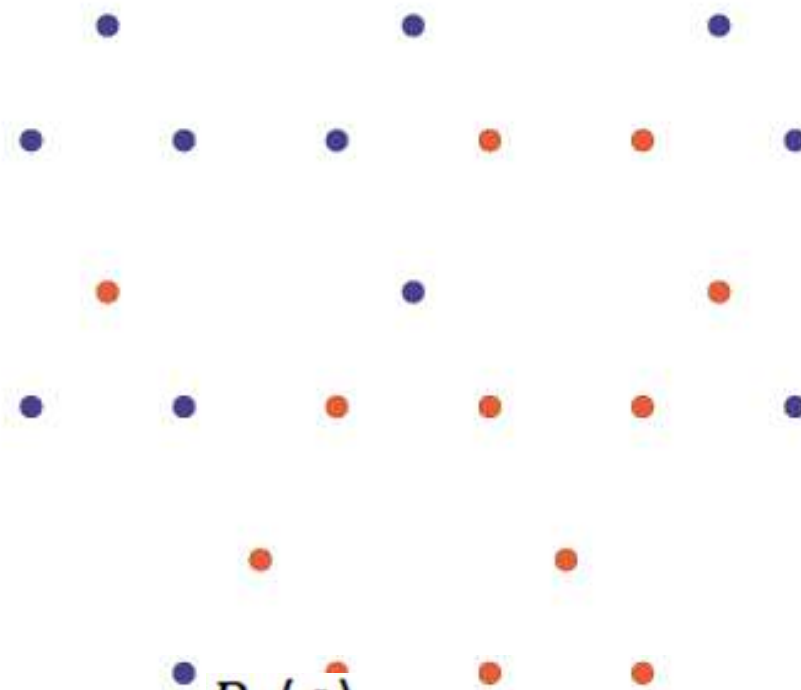
3 Vertices and 0 Edges



★ λ ways to color independently each of the vertices $|u, v, w$.

$$P_3(\lambda) = \lambda^3$$

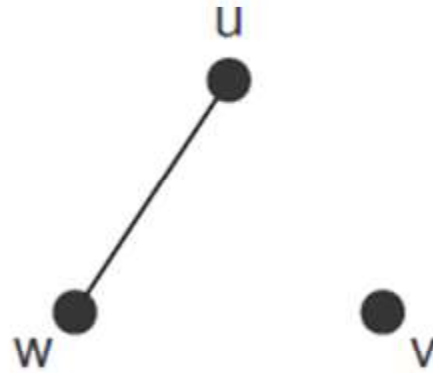
3 Vertices, 0 Edges, and 2 colors



$P_3(2)$

$$P_G(2) = 2^3 = 8$$

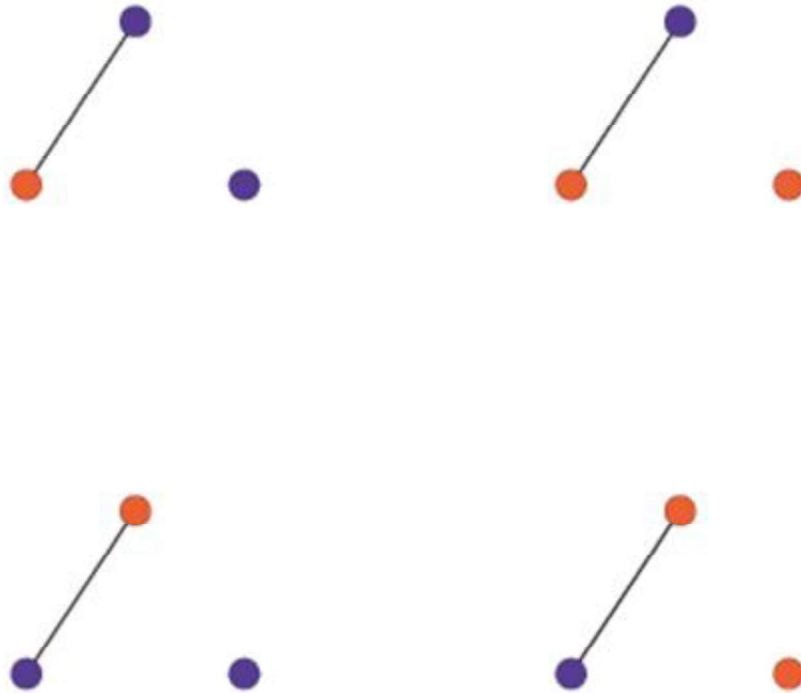
3 Vertices and 1 Edge



- ★ λ ways to color v ; λ ways to color u ; $\lambda - 1$ ways to color w that cannot get the color of u .

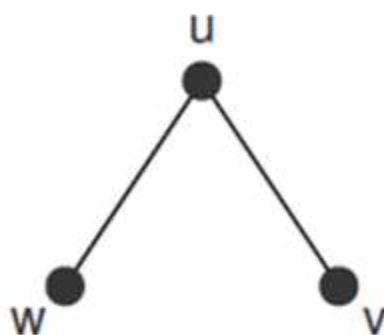
$$\begin{aligned} P_3(\lambda) &= \lambda^2(\lambda - 1) \\ &= \lambda^3 - \lambda^2 \end{aligned}$$

3 Vertices, 1 Edge, and 2 colors



$$P_3(2) = 2^3 - 2^2 = 4$$

3 Vertices and 2 Edges



λ ways to color u ; $\lambda - 1$ ways to color v that cannot get the color of u ; $\lambda - 1$ ways to color w that cannot get the color of u .

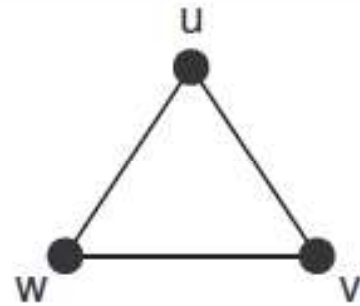
$$\begin{aligned} P_3(\lambda) &= \lambda(\lambda - 1)^2 \\ &= \lambda^3 - 2\lambda^2 + \lambda \end{aligned}$$

3 Vertices, 2 Edges, and 2 colors



$$P_3(2) = 2^3 - 2 \cdot 2^2 + 2 = 2$$

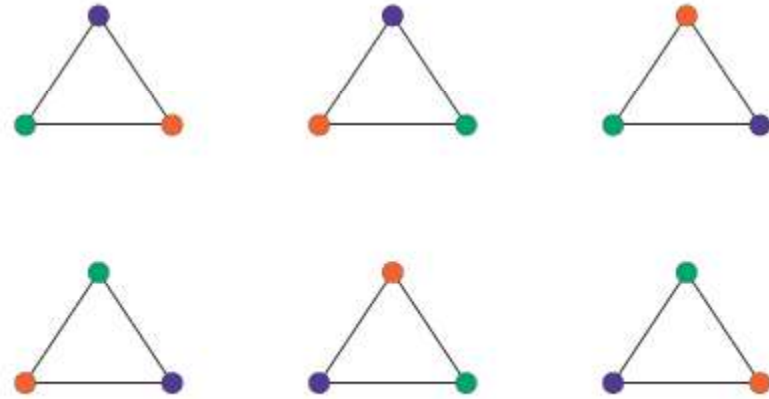
3 Vertices and 3 Edges



λ ways to color u ; $\lambda - 1$ ways to color v that cannot get the color of u ; $\lambda - 2$ ways to color w that cannot get the colors of u and v .

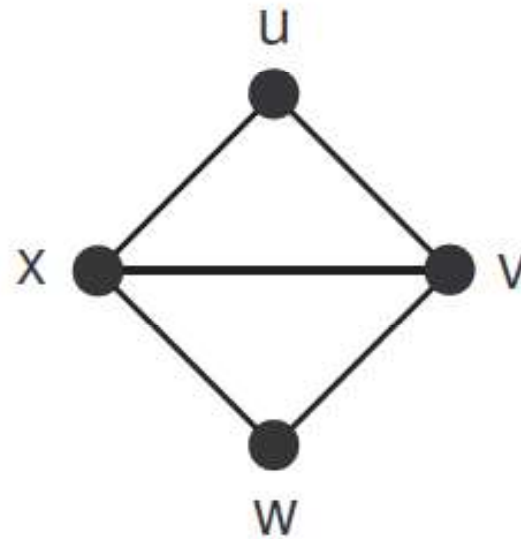
$$\begin{aligned}P_3(\lambda) &= \lambda(\lambda - 1)(\lambda - 2) \\&= \lambda^3 - 3\lambda^2 + 2\lambda \\&= (\lambda - 1)^3 - (\lambda - 1)\end{aligned}$$

3 Vertices, 3 Edges, and 3 colors



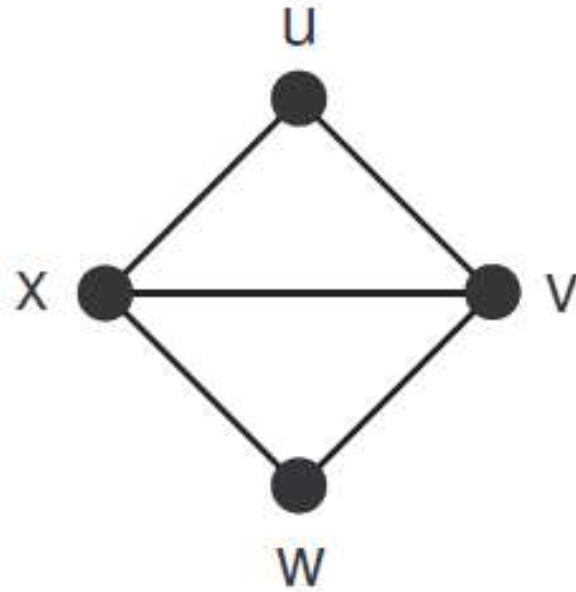
$$P_3(3) = 3 \cdot 2 \cdot 1 = 6$$

4 Vertices and 5 Edges



- ★ λ ways to color v ; $\lambda - 1$ ways to color x that cannot get the color of v ; $\lambda - 2$ ways to color u that cannot get the colors of v and x ; $\lambda - 2$ ways to color w that cannot get the colors of v and x .

4 Vertices and 5 Edges



$$\begin{aligned}P_4(\lambda) &= \lambda(\lambda-1)(\lambda-2)^2 \\ &= \lambda^4 - 5\lambda^3 + 8\lambda^2 - 4\lambda\end{aligned}$$

Null Graphs – N_n



- ★ The null graph N_n has n vertices and no edges.
- ★ Each vertex can be colored independently with k colors.

$$\begin{aligned} P_n(\lambda) &= \lambda^n \\ &= \lambda^n - 0 \cdot \lambda^{n-1} + 0 \cdot \lambda^{n-2} - \dots + 0 \end{aligned}$$

Stars – S_n



- ★ The Star graph S_n has $n - 1$ edges. A root vertex is connected to the rest of the $n - 1$ vertices each connected only to the root.
- ★ The root can be colored with λ colors and each of the other $n - 1$ vertices can be colored with $\lambda - 1$ colors.

$$\begin{aligned} P_n(\lambda) &= \lambda(\lambda - 1)^{n-1} \\ &= \lambda^n - (n - 1)\lambda^{n-1} + \dots + 0 \end{aligned}$$

Theorem: A graph of n vertices is complete graph if and only if its chromatic polynomial is

$$P_n(\lambda) = \lambda (\lambda - 1)(\lambda - 2) \dots (\lambda - (n-1))$$

Proof: Let given graph is complete graph.

Choose any vertex and paint with colors. The number of ways of different coloring for the selected vertex is λ .

Now choose second vertex which is adjacent to first vertex and it can be properly colored in $(\lambda - 1)$ ways.

Same way the n^{th} vertex will be coloured in $(\lambda - (n-1))$ ways if and only if every vertex is adjacent to every other. That is if graph is complete.

And chromatic polynomial is

$$P_n(\lambda) = \lambda (\lambda - 1)(\lambda - 2) \dots (\lambda - (n-1))$$

Theorem: An n -vertex graph is a tree iff its chromatic polynomial

$$P_n(\lambda) = \lambda (\lambda - 1)^{n-1}$$

Proof: Let G be a tree

To prove $P_n(\lambda) = \lambda (\lambda - 1)^{n-1}$

Using induction on n ,

For $n = 1$, the result is trivial

i.e. $P_1(\lambda) = \lambda$

Assume the result is true for a tree with at most $(n - 1)$ vertices, where $n \geq 2$.

i.e. $P_{n-1}(\lambda) = \lambda (\lambda - 1)^{n-2}$.

Let G be a tree with n vertices and e be the pendant edge of G .

$$\begin{aligned} \text{Now } P_n(\lambda) &= P_{n-1}(\lambda)P_1(\lambda) \\ &= \lambda (\lambda - 1)^{n-1}. \end{aligned}$$

Conversely, Let G be a simple graph with chromatic polynomial

$$P_n(\lambda) = \lambda(\lambda - 1)^{n-1}$$

$$= \lambda(\lambda^{n-1} - (n-1)\lambda^{n-2} + \dots + (-1)^{n-1})$$

$$= \lambda^n - (n-1)\lambda^{n-1} + \dots + (-1)^n \lambda$$

which is monic polynomial with n vertices.

The highest degree is n .

Also the last term $(-1)^n \lambda$ ensures that G is connected.

The next highest degree is $(n-1)$ and coefficient of $|\lambda^{n-1}|$ is $(n-1)$, which gives number of edges in the graph.

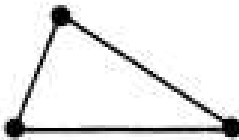
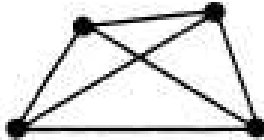
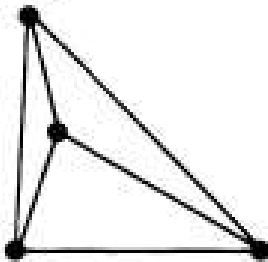
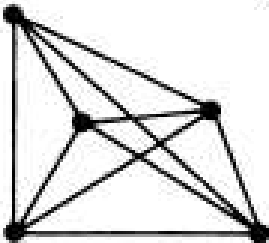
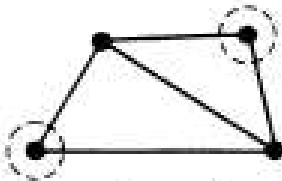
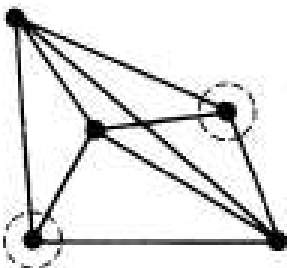
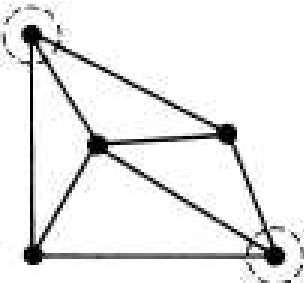
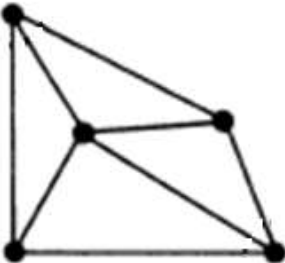
i.e. G is a Tree.

THEOREM

Let a and b be two nonadjacent vertices in a graph G . Let G' be a graph obtained by adding an edge between a and b . Let G'' be a simple graph obtained from G by fusing the vertices a and b together and replacing sets of parallel edges with single edges. Then

$$P_n(\lambda) \text{ of } G = P_n(\lambda) \text{ of } G' + P_{n-1}(\lambda) \text{ of } G''.$$

Ex: Find Chromatic Polynomial of Given graph.



$$\lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)$$

$$\lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)$$

$$\lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)$$

$$\lambda(\lambda - 1)(\lambda - 2)$$

$$\begin{aligned}
 P_5(\lambda) \text{ of } G &= \lambda(\lambda - 1)(\lambda - 2) + 2\lambda(\lambda - 1)(\lambda - 2)(\lambda - 3) \\
 &\quad + \lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4) \\
 &= \lambda(\lambda - 1)(\lambda - 2)(\lambda^2 - 5\lambda + 7)
 \end{aligned}$$

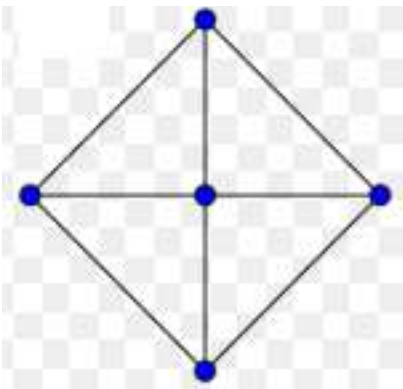
Evaluation of a chromatic polynomial.

Ex: Find Chromatic Polynomial of Given graph.

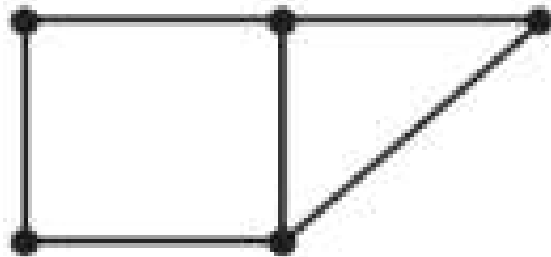
1.



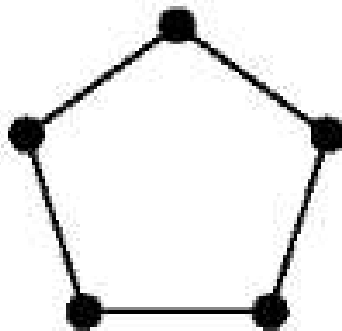
2.



3.



4.



Matching:

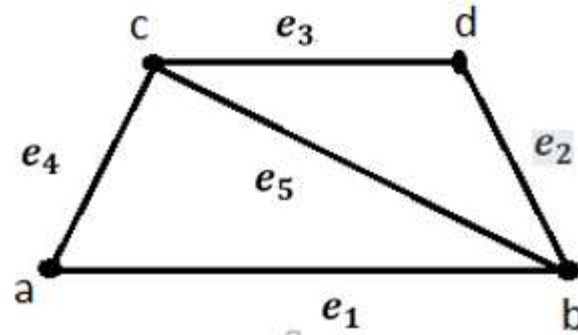
Matching in a graph is subset of edges in which no two edges are adjacent.

Remark: A single edge in a graph is obviously a matching.

Example:

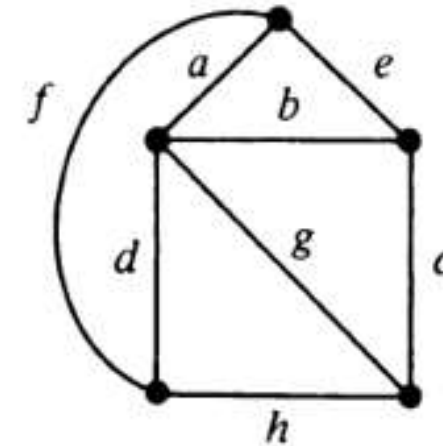
Matching are $\{e_1\}$, $\{e_2\}$, $\{e_3\}$, $\{e_4\}$, $\{e_5\}$,

$\{e_1, e_3\}$, $\{e_2, e_3\}$

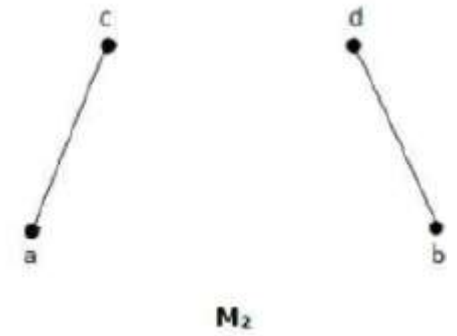
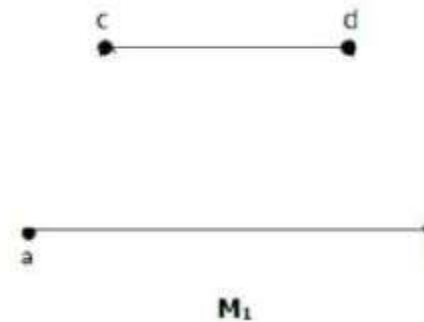
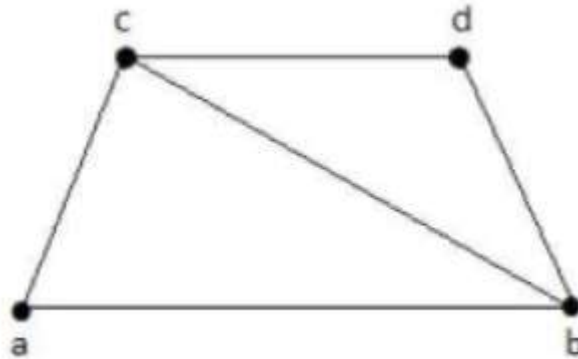


Example: In the given graph the matchings are

$M_1 = \{h, b\}$, $M_2 = \{a, c, h\}$, $M_3 = \{e, g, d, h\}$, $M_4 = \{c, d\}$, $M_5 = \{a, c\}$ etc .



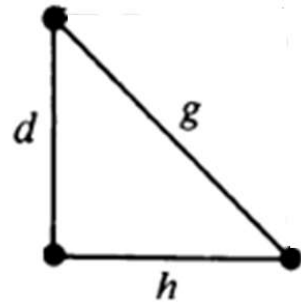
Example:



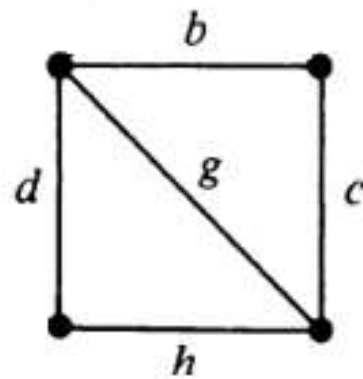
Maximal Matching:

A matching of the graph is said to be maximal in which no other edges can be added.

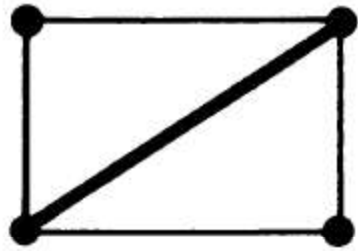
Ex1 : In a complete graph of three vertices any single edge is a matching.
i.e. maximal matching are $\{d\}$, $\{h\}$, $\{g\}$.



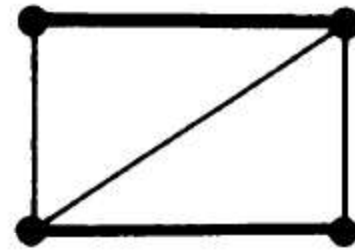
Ex 2 : In the given graph the maximal matching are $\{b, h\}$, $\{d, c\}$, $\{g\}$ is the maximal matching.



Ex 3 : Graph and its two maximal matching are



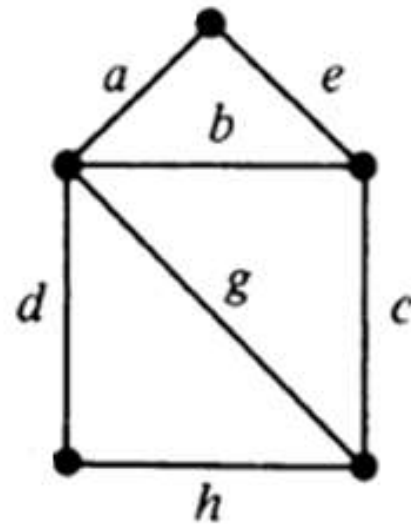
(a)



(b)

Graph and two of its maximal matchings.

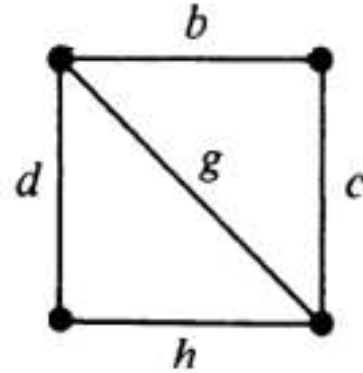
Ex: $\{h, b\}$, $\{d, c\}$, $\{d, e\}$, $\{c, a\}$, $\{g, e\}$, $\{h, a\}$, $\{h, e\}$ are maximal matchings of given graph.



Largest Maximal Matching or Maximum Matching:

A maximal matching of the graph with largest number of edges is defined as largest matching.

Ex: In the given graph the largest matchings are $\{b, h\}, \{d, c\}$.



Matching Number:

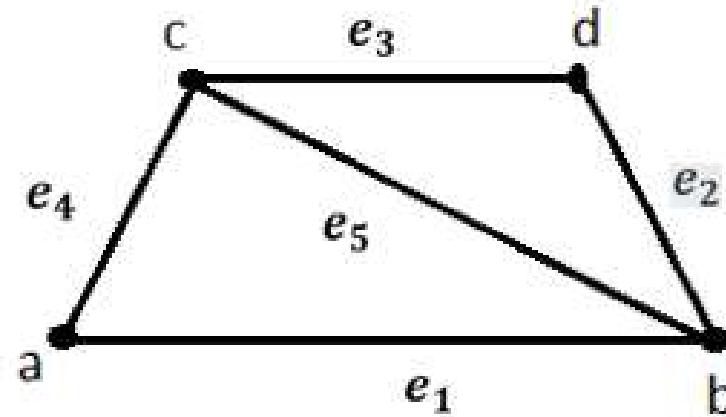
The number of edges in largest matching of G is called matching number.

Ex: In above example the matching number is 2.

Perfect Matching or Complete Matching:

A matching M of a graph G is said to be a perfect match if every vertex of G is incident to exactly one edge of the matching M .

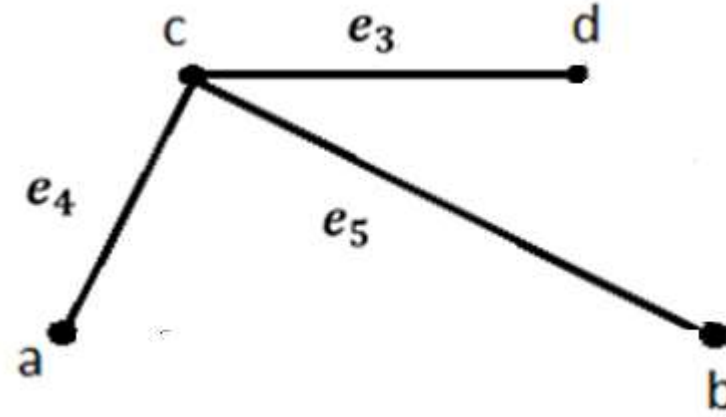
Example: $M_1 = \{e_1, e_3\}$, $M_2 = \{e_2, e_4\}$ are perfect matchings of G .



Note: Every perfect matching of graph is also a maximum matching of a graph, because there is no chance of adding one more edge in a perfect matching graph. But a maximum matching need not to be a perfect matching.

Note: If a graph G has perfect matching then number of vertices is always even but the converse need not be true.

$M_1 = \{e_4\}, M_2 = \{e_5\}, M_3 = \{e_3\}$ are maximal matching of given graph but not perfect matching Though it has even number of vertices.



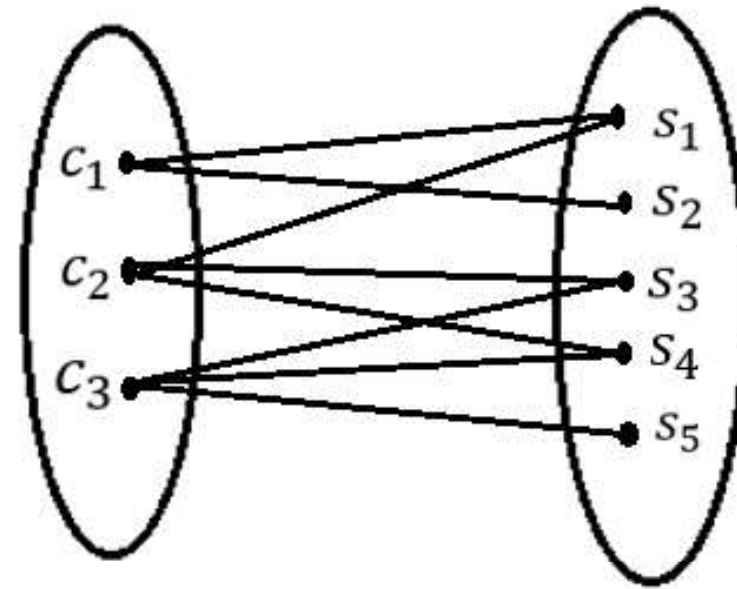
Note: If the number of vertices in the graph is odd then perfect matching does not exist but the converse need not to be true.

Complete matching in a Bipartite graph:

Theorem: A complete matching of V_1 into V_2 in a bipartite graph exist if and only if every subset of r vertices in V_1 is collectively adjacent to r or more vertices V_2 for all values of r .

Example:

Complete matching in bipartite graph.



Covering

Edge Covering or line Covering:

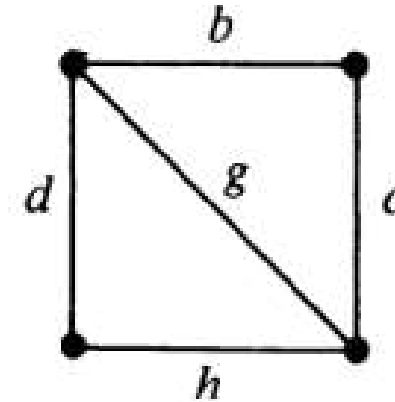
In a graph G a set of edges g is said to be cover G if every vertex in G is incident on at least one edge in g .

A set of edges that covers a graph G is said to be an edge covering or cover

Ex: A graph G is trivially its own covering.

A spanning tree in a connected graph is another covering.

A Hamiltonian circuit in a graph is also a covering(if it exists).

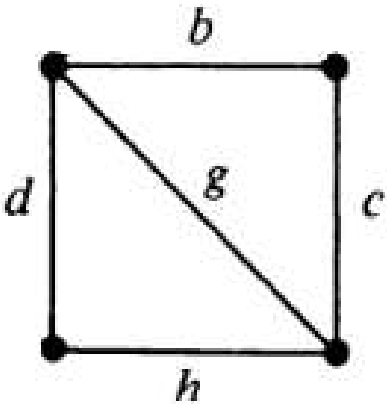


Ex: Edge covering are $\{b, g, h\}, \{d, c\}, \{b, h\}, \{b, d, h\}, \{b, c, h\}, \{d, b, c\}$ etc

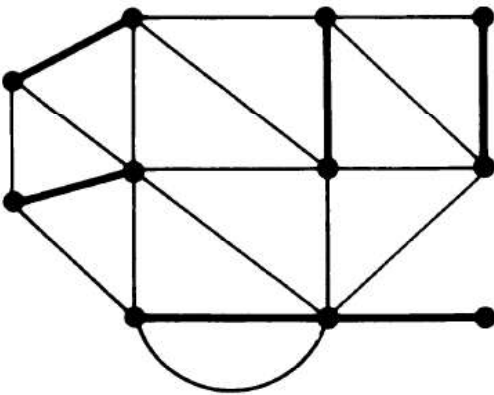
Minimal edge covering:

An **edge covering** of a graph G is said to be minimal if no edge can be removed without destroying its ability to cover the graph.

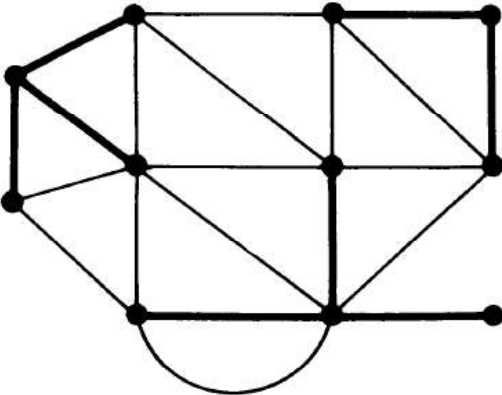
Ex: $\{b, h\}, \{d, c\}, \{b, d, g\}$ are minimal line covering of G .



Ex:



(a)



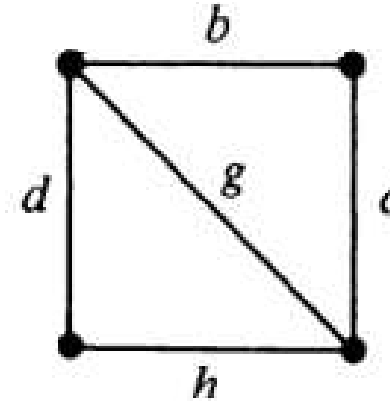
(b)

Graph and two of its minimal coverings.

Minimum Edge covering:

A minimal line covering with minimum number of edges is called minimum line covering.

Ex: $\{b, h\}, \{d, c\}$



Edge covering number:

A graph, in general has many edge covering, and they may be of different sizes.(consisting of different number of edges)

The number of edges in a minimum line covering in G is called the line covering number of G .
covering number =2 for given graph G .

- A covering exists for a graph if and only if the graph has no isolated vertex.
- A covering of n -vertex graph will have at least $\lceil n/2 \rceil$ edges. ($\lceil x \rceil$ denotes the smallest integer not less than x) .
- Every line covering contains a minimal covering.
- No minimal covering can contain a circuit for we can always remove an edge from a circuit without leaving any of vertices in the circuit uncovered. Therefore, a minimal covering of an n -vertex graph can contain no more than $n - 1$ edges.

Theorem:

A covering $\mathcal{g}$ of a graph G is minimal if and only if $\mathcal{g}$ contains no path of length three or more.

Proof: Let $\mathcal{g}$ be a minimal covering of a graph and we will prove that $\mathcal{g}$ contains no path of length three or more.

Let $\mathcal{g}$ contains a path of length three and it is $V_1 e_1 V_2 e_2 V_3 e_3 V_4$.

Edge e_2 can be removed without leaving its end vertices v_2 and v_3 uncovered, still $\mathcal{g}$ is a covering, which is a contradiction that $\mathcal{g}$ is a minimal covering. Hence $\mathcal{g}$ contains no path of length three or more.

Conversely, let a covering $\mathcal{g}$ contains no path of length three or more, therefore all its component must be a star graph (i.e., graphs in the shape of stars). From a star graph no edge can be removed without leaving a vertex uncovered, hence, $\mathcal{g}$ must be a minimal covering.

Four color problem

So far we have considered proper coloring of vertices and proper coloring of edges. Now, let us consider the proper coloring of regions in a planar graph.

The regions of the planar graph are said to be properly colored if no two contiguous or adjacent regions have the same color. (Two regions are said to be adjacent if they have a common edge between them.)

The proper coloring of the regions is also called map coloring.

Example:

In atlas different countries are colored such that countries with common boundaries are shown in different colors.

Four color conjecture:

To color the regions of the graph with the minimum number of colors leads us to the most famous conjecture in graph theory which stated that every map (i.e., a planar graph) can be properly colored with four colors.